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B.Tech. / M.Tech. (Integrated) DEGREE EXAMINATION, DECEMBER 2025

First Semester

21BTB103T - BIOLOGY

(For the candidates admitted during the academic year 2025 - 2026)

Note:

- i. **Part - A** should be answered in OMR sheet within first 40 minutes and OMR sheet should be handed over to hall invigilator at the end of 40th minute.
- ii. **Part - B** and **Part - C** should be answered in answer booklet.

Time: 3 hours

Max. Marks: 75

PART - A (20 × 1 = 20 Marks)

Answer **all** Questions

Marks CO BL

- | | | | |
|---|---|---|---|
| 1. The cell division ----- is for growth and maintenance.
(A) mitosis
(B) Meiosis
(C) Cytokinensis
(D) Interphase | 1 | 1 | 1 |
| 2. ----- is the protein that makes up nails and hair.
(A) Keratin
(B) albumin
(C) aminoacids
(D) biotin | 1 | 1 | 1 |
| 3. The oxidation of fatty acids takes place in
(A) Mitochondria
(B) nucleus
(C) nucleolus
(D) cell wall | 1 | 1 | 1 |
| 4. COVID 19 Vaccines
(A) Whole cell
(B) DNA particle
(C) Attenuated cells
(D) Protein | 1 | 1 | 1 |
| 5. RNA differs DNA by the base -----
(A) adenine
(B) uracil
(C) bond
(D) thymine | 1 | 2 | 2 |
| 6. Nucleic acids are chains of 5-carbon sugars linked by which type of bonds with an organic base protruding from each sugar?
(A) amino
(B) hydrogen
(C) oxygen
(D) phospho diester bonds | 1 | 2 | 4 |
| 7. Find out the polysaccharide
(A) Starch
(B) Sucrose
(C) glucose
(D) fructose | 1 | 2 | 2 |
| 8. Which type of fatty acid has a single bond between the carbons?
(A) saturated
(B) unsaturated
(C) alkaline
(D) monounsaturated | 1 | 2 | 1 |
| 9. -----selectively recognize proper substrates over the molecules
(A) proteins
(B) `sugar
(C) Enzymes
(D) glucose | 1 | 3 | 5 |
| 10. The pigment which is responsible for absorption of light is -----
(A) xylem
(B) chloroplast
(C) chlorophyll
(D) phloem | 1 | 3 | 1 |

11. The source of glucose for living organism is ----- (A) carbohydrates (B) proteins (C) fat (D) oils	1	3	2
12. When the temperature increases, protein will (A) denature (B) grow (C) doubles (D) shrinks	1	3	1
13. Microfilaments are made of (A) actin (B) fibrin (C) tubulin (D) cilia	1	4	2
14. Kinesin and dynein transport ----- along microtubule (A) actin (B) cargo (C) myosin (D) atp	1	4	3
15. Catalytic site present in ATP synthase is ----- (A) F1 subunit (B) outer membrane (C) inner membrane (D) F0	1	4	2
16. Flagella is made up of ----- (A) microtubules (B) flagellin (C) proteins and lipids (D) thiamin	1	4	1
17. Which of the following biomaterial is used by cardiologist (A) Titanium (B) Ceramic (C) Poly carbonate (D) Microplastic	1	5	3
18. Which of the following property should be present in a bio material used by ortho surgeons (A) load bearing capacity (B) biodegradable (C) Conduct electricity (D) mimic blood cells	1	5	1
19. Ceramic is used in (A) Dental implants (B) stent (C) bone replacement (D) hair oil	1	5	2
20. Bio mimetic is designed to (A) using micro machines in life science application (B) replacement of machines using bio material (C) designing using biological concepts (D) using machines instead of biologic material	1	5	1

PART - B (4 × 10 = 40 Marks)

Answer any 4 Questions

	Marks	CO	BL
21. Give an account of the Plant cell with a diagram	10	1	3
22. Explain the structure of DNA	10	1	3
23. Explain carbohydrates and their classes	10	2	3
24. Describe the resistance mechanism of bacteria against the antibiotics	10	3	3
25. Explain the mechanism of flagellar motor with diagrams.	10	4	3
26. List out the properties of material that can be used in medical application	10	5	3

PART - C (1 × 15 = 15 Marks)

Answer **any 1** Question

Marks CO BL

27. Compare and contrast the mechanism of action of differed vaccines.

15 3 4

28. Explain mitosis and meiosis with necessary diagrams.

15 2 4

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